

TrES-3b

R-2773

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140626 · created 2026-09-03T23:19:46+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456834.85491 +/- 0.00057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.016
Transit depth (Rp/Rs)^2	3.83 +/- 0.63 [%]
Orbital Inclination (inc)	81.51 +/- 0.33
Airmass coefficient 1 (a1)	1.038 +/- 0.013
Airmass coefficient 2 (a2)	-0.031 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	#1 - [547, 157]
Optimal Aperture	4.02
Optimal Annulus	7.74
Transit Duration (day)	0.0555 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.016 vs 0.16309 (deviation 1.17σ)
Duration fit vs published	0.0555 d vs 0.0567 d (deviation 0.18σ)
Mid-time O–C	-3.6 min
Depth SNR	7.0
Residual scatter	1.28 %

Light curve

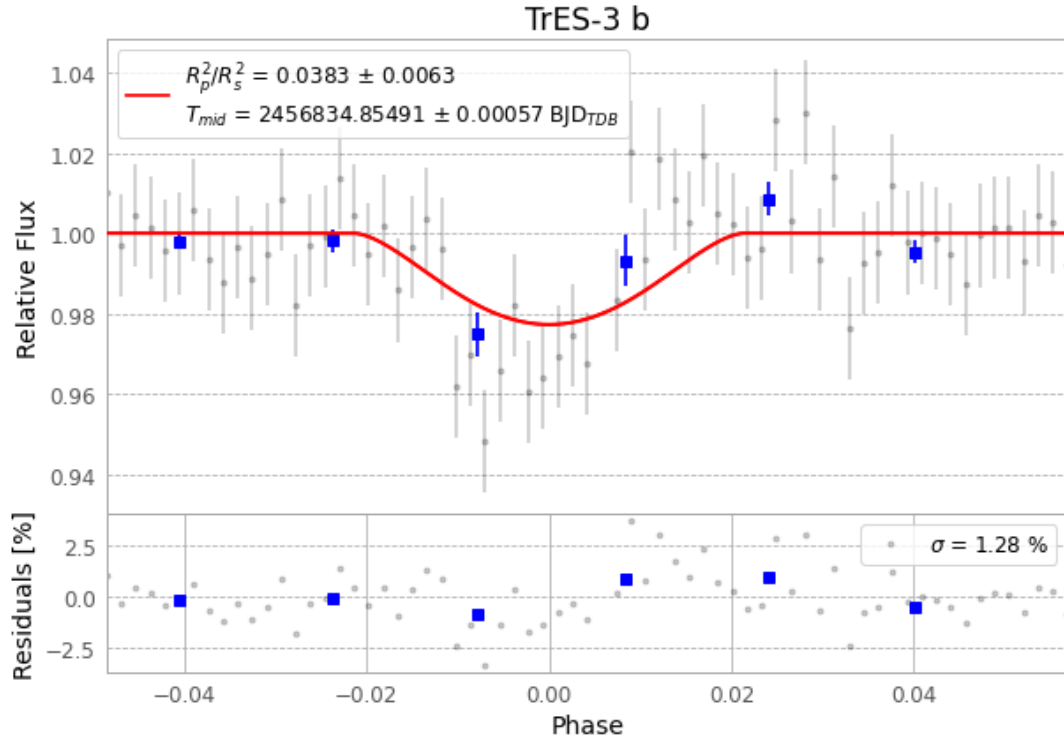


Figure 1 — FinalLightCurve_TrES-3 b_2014-06-25.png (SHA-256 7cdb1c21a41f2243...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e9364becc78b3d0f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3140626065416.FITS → TRES-3140626101212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140626.log (339c6c8d8b96...), FinalLightCurve_TrES-3 b_2014-06-25.png (7cdb1c21a41f...), FinalLightCurve_TrES-3 b_2014-06-25.csv (53d58ee2cc3f...)

Compute resources

Wall time 151.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 23:19Z.